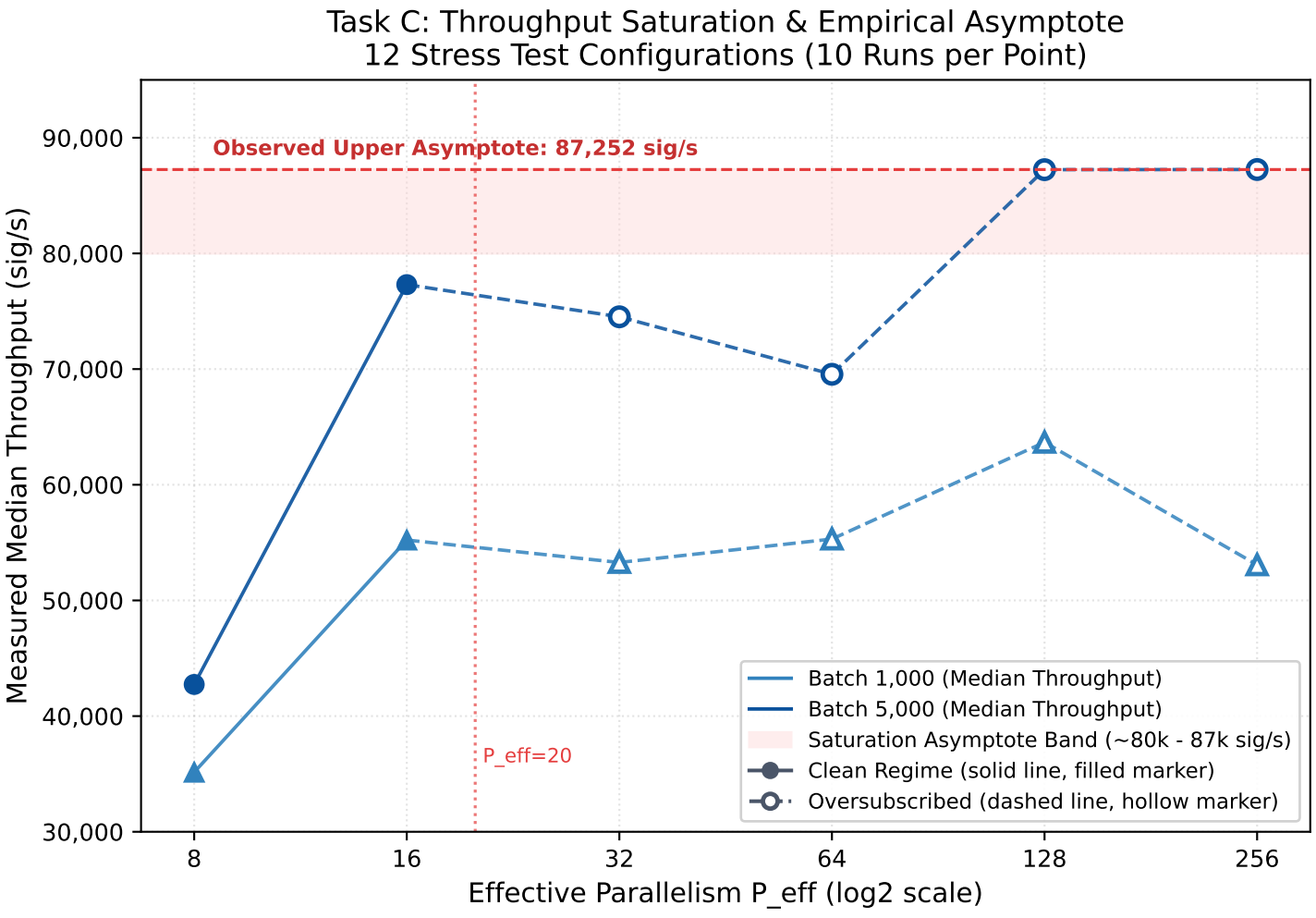
Task C: Throughput Saturation & Empirical Asymptote
12 Stress Test Configurations (10 Runs per Point)
Observed Upper Asymptote: 87,252 sig/s
P_eff=20
Measured Median Throughput (sig/s)
Effective Parallelism P_eff (log2 scale)
90,000
80,000
70,000
60,000
50,000
40,000
30,000
8
16
32
64
128
256
Batch 1,000 (Median Throughput)
Batch 5,000 (Median Throughput)
Saturation Asymptote Band (~80k - 87k sig/s)
Clean Regime (solid line, filled marker)
Oversubscribed (dashed line, hollow marker)